

L2 - Graphing Sine and Cosine Functions

MCR3U

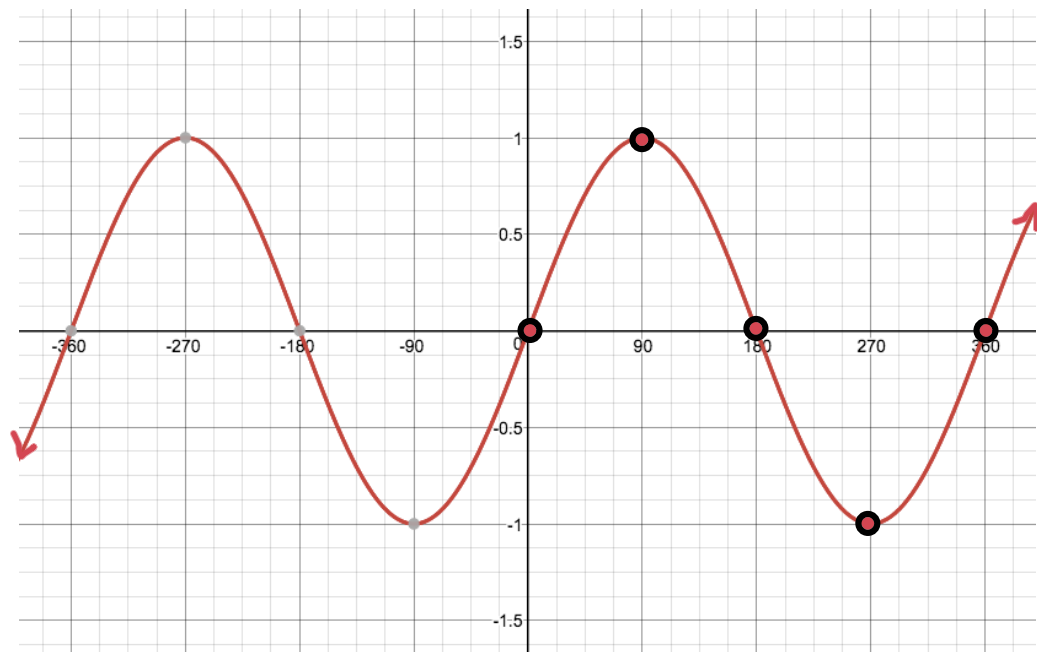
Section 1: Graphing Sine and Cosine

[DESMOS demonstration](#)

To graph sine and cosine, we will be using a Cartesian plane that has angles for x values.

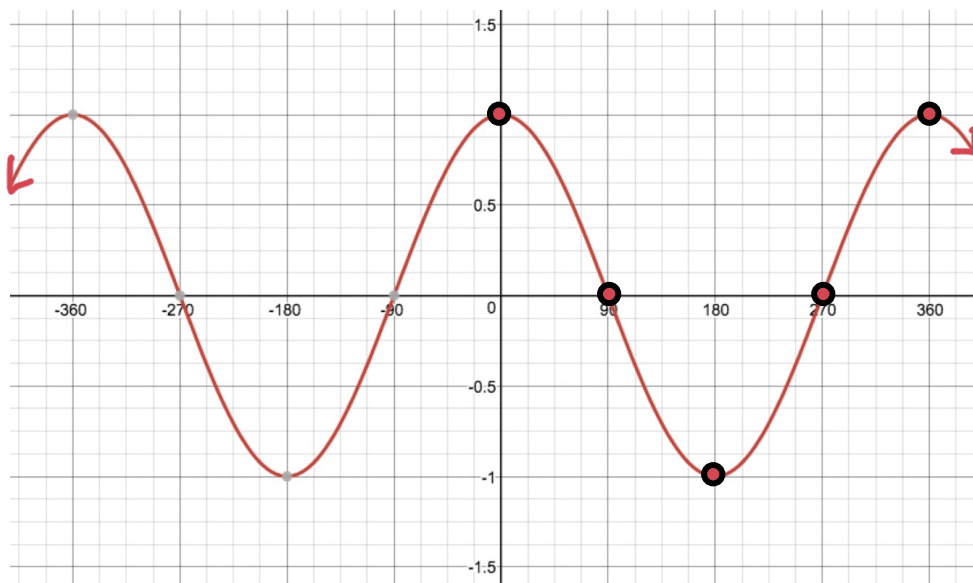
Example 1: Complete the following table of values for the function $f(x) = \sin(x)$. Use special triangles, the unit circle, or a calculator to find values for the function at 30° intervals. Use the table to graph the function.

x	$f(x)$
0	0
30	0.5
60	$\frac{\sqrt{3}}{2} \sim 0.87$
90	1
120	$\frac{\sqrt{3}}{2} \sim 0.87$
150	0.5
180	0
210	-0.5
240	$-\frac{\sqrt{3}}{2} \sim -0.87$
270	-1
300	$-\frac{\sqrt{3}}{2} \sim -0.87$
330	-0.5
360	0



Example 2: Complete the following table of values for the function $f(x) = \cos(x)$. Use special triangles, the unit circle, or a calculator to find values for the function at 30° intervals. Use the table to graph the function.

x	$f(x)$
0	1
30	$\frac{\sqrt{3}}{2} \sim 0.87$
60	0.5
90	0
120	-0.5
150	$-\frac{\sqrt{3}}{2} \sim -0.87$
180	-1
210	$-\frac{\sqrt{3}}{2} \sim -0.87$
240	-0.5
270	0
300	0.5
330	$\frac{\sqrt{3}}{2} \sim 0.87$
360	1



Section 2: Properties of Sine and Cosine Functions

Domain: $\{X \in \mathbb{R}\}$

Range: $\{Y \in \mathbb{R} \mid -1 \leq y \leq 1\}$

Period: 360°

Amplitude: $\frac{\text{max}-\text{min}}{2} = \frac{1-(-1)}{2} = 1 \text{ unit}$

Section 3: Transformations of the Sine and Cosine Functions

$$y = a \sin[k(x - d)] + c$$

[Desmos Demonstration](#)

a	k	d	c
Vertical stretch or compression by a factor of a .	Horizontal stretch or compression by a factor of $\frac{1}{k}$.	Phase shift $d > 0$; <i>shift right</i>	Vertical shift $c > 0$; <i>shift up</i>
Vertical reflection if $a < 0$	Horizontal reflection if $k < 0$.	$d < 0$; <i>shift left</i>	$c < 0$; <i>shift down</i>
$ a = \text{amplitude}$	$\frac{360}{ k } = \text{period}$		

Example 3: For the function $y = 3 \sin[2(\theta + 60^\circ)] - 1$, state the...

Amplitude: $a = 3$	Period: $\text{period} = \frac{360}{ k } = \frac{360}{2} = 180^\circ$
Phase shift: $d = -60^\circ$; Shift left 60°	Vertical shift: $c = -1$; Shift down 1 unit
Max: $\text{max} = c + a = -1 + 3 = 2 \text{ units}$	Min: $\text{min} = c - a = -1 - 3 = -4 \text{ units}$